

SCI-FLT-FIXED v0.1 Shared Normative Scientific Core

SCI-FLT-FIXED-NORMATIVE-CORE v0.1/freeze-candidate

Status: implementation-blind conditional scientific-owner freeze candidate; owner signature required

Scientific owner: Grant Wilson

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Author packet manifest SHA-256: 7f2d03f182258ac9770f7dba869e9ae0b5018efdcdb93b18b299a9b9c6df1e4d
r0.2 owner directive SHA-256: 02ca63dcc8ac19e554fa333c20773ff85264336b4f622024605f6514ec1716d1
r0.3 owner directive SHA-256: c1de5ad5b9f9e5451ec2a3d9d9c18d30f7990eec721497c9d473aaeb2b043551
r0.4 owner directive SHA-256: 855d9709d3fe3b4d71b6a56a020e52ce52164a0384a2ed2d562145f8fb3b0a99
final owner directive SHA-256: c30c49ca19ec15b384ed8178de233e473513fdc520e090c152f571a8a802bf0c
Build recipe SHA-256: 19d4ac5b4238d704a160528f2dc52831ed944cdd48a3f98f1764509ef066ac53
```

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1. Authority, scope, and normative language

This document is the sole shared normative scientific core for the SCI-FLT-FIXED v0.1 conditional freeze candidate. It incorporates the exact r0.2 formal-closure directive, the exact r0.3 final targeted scientific-closure and source-preflight directive, the exact r0.4 final formal-closure and freeze-preflight directive, and the exact final micro-repair and conditional-freeze directive bound by the Stage B build record. It preserves the scientific authority of the exact 17-object packet bound by AUTHOR_PACKET_MANIFEST.md identity SCI-FLT-FIXED_AUTHOR_PACKET v0.1/r0.1. The external SHA-256 of that manifest is 7f2d03f182258ac9770f7dba869e9ae0b5018efdcdb93b18b299a9b9c6df1e4d.

The key words MUST, MUST NOT, REQUIRED, SHALL, SHALL NOT, SHOULD, SHOULD NOT, MAY, and OPTIONAL state the strength of a scientific-contract requirement. Unknown, unavailable, disabled, failed, and zero are distinct typed states.

SCI-FLT is the recovery tranche. SCI-FLT-FIXED is the v0.1 package. No SCI-FLT-INF contract is defined here.

2. Scientific object and estimand

Define the exact parent fact domain and the scientific numerical signal domain separately:

```
S_parent_fact
  = exact parent row-identity and fact domain;

D_m
  = {q in S_parent_fact:
      an available finite real signal payload exists at q};

m : D_m -> R.
```

A missing, unavailable, or non-finite stored payload is a typed row fact in S_parent_fact ; it is not an element of the scientific numerical function m . Parent-shaped storage is a representation containing payload and state records, not the scientific vector domain. The transformation MUST establish q in D_m before evaluating m_q .

For one exact admitted parent, the scientific object is

```
J_full = J(plan,
    immutable S_parent_fact and row membership,
    FLT input admission,
    typed availability and finiteness facts defining D_m,
    support,
    required predicates).

A_Theta,J = J_full L_Theta,
y = A_Theta,J m,
y : S_out -> R.
```

The base scalar field is real on its typed function domains:

```
m : D_m -> R; y : S_out -> R;
k_Theta(r), L_Theta, and A_Theta,J are real-valued.
```

Every admitted sampled coefficient is finite, real, unit-typed, present in one canonical exact representation, and content-bound before application. A missing, non-finite, complex, unrepresentable, or conflicting

coefficient makes plan resolution unavailable. The prospective numerical-comparison policy cannot repair or admit such a coefficient.

The only numerically admitted base family is FLT-FIXED-CONV. L_{Theta} is the complete matrix representation of one exact realized sampled convolution; it is not a separately selectable arbitrary dense linear-operator family. FLT-FIXED-CONV-LOWPASS is only a qualified subtype of FLT-FIXED-CONV.

L_{Theta} and the externally resolved plan are complete and frozen before application to the parent random field. Parent payload amplitudes do not select, learn, tune, or alter any plan or operator fact. J_{full} is resolved once from only declared immutable parent identity and row membership, exact parent-row admission, declared typed row-state facts defining D_m , support, and exact required predicates. Reading those row-state facts is structural screening, not evaluation of m_q , convolution arithmetic, or permission to tune the plan. The resulting A_{Theta}, J is the complete applied operator. The output y is transformed parent-map amplitude on the selected row domain. There is no additive term.

Strict linearity is conditional on the exact frozen parent membership and J_{full} . It is not a global-linearity claim across parents whose domain, support, admission, availability, finiteness, or validity differs. Response, covariance, noise realizations, and NOI members receive the identical frozen A_{Theta}, J ; J_{full} is never re-resolved for a response perturbation, covariance draw, noise realization, or fixed-state NOI member. A member that cannot supply one frozen required footprint is unavailable on the affected row rather than receiving another selector. Selection and support uncertainty are excluded from conditional linear propagation unless separately supplied as a typed uncertainty object.

Fixed convolution is the admitted structured family. Fixed low-pass convolution is only a qualified subtype of fixed convolution. It is not a second generic family and it is not implied by a friendly name, a width, or the word smoothing.

3. Parent roles and parent ordering

One transformation binds exactly one immutable parent in exactly one role. The exact signal-role disposition is:

Parent role	Exact m used by FLT-SIG
FLT-PARENT-MAP-OBS	base/unfiltered MAP observation signal
FLT-PARENT-MAP-COADD	base/unfiltered MAP coadd signal
FLT-PARENT-JINC-OBS	normalized $jinc_{\text{map}}$ on admitted local support

The roles are not interchangeable even when shape, WCS values, or numerical payloads happen to agree. MAP observation, MAP coadd, and JINC observation successors remain separately identified. SCI-FLT-FIXED does not coadd and does not presume that filtering and coaddition commute.

The exact parent binding carries the complete parent package, revision, product and application generations, role, membership, stable array or group, quantity, units and originating nominal-beam convention, WCS, frame, topology, grid, pixel metric, shape, row domain, support, validity and causes, response and covariance availability, null and additive-reference state, exposure, causal influence, lifecycle, failure, and provenance facts required by the applicable MAP or JINC boundary. For JINC, all five required numerical roles form one atomic parent:

```
jinc_signal_numerator
jinc_signed_normalization
jinc_quadratic_accumulator
jinc_map with local support/validity
jinc_coefficient_squared_time
```

Only $jinc_{\text{map}}$ is the JINC signal vector m . The numerator, signed normalization, quadratic accumulator, and coefficient-squared temporal accounting are diagnostics or parent facts, not alternate FLT signals.

Only the selected signal role receives L_{Theta} as FLT-SIG. Parent response receives its separately typed response composition and parent covariance its separately typed covariance composition. Support and validity are predicates and records, not convolved signals. Exposure is lineage and is not convolved into

physical exposure. Normalization and coefficient diagnostics remain parent facts. No other map-shaped role is transformed without a separately named method. A partial five-role JINC bundle is not a parent.

The Stage A boundary records make ordinary numerical MAP and ToITEC JINC parents unavailable at this draft's launch state. The contract defines their typed successor routes but does not manufacture numerical parents.

4. Requested, effective, resolved, and applied state

The contract distinguishes:

- requested scientific purpose and method;
- effective selection or disablement;
- complete externally resolved immutable plan;
- observation-resolved operator generation;
- exact frozen row selector and applied transformation;
- complete_publication_disposition_candidate with its exact variant;
- publication decision; and
- realized atomic successor product.

Every coefficient, kernel parameter, cutoff or width, WCS fact, normalization, support rule, transfer qualification, threshold, and lifecycle fact is resolved before application under one exact FLT-owned plan and fixed before use. "Externally resolved" describes timing and separation from payload arithmetic; it does not transfer FLT policy or transformation ownership to an unnamed producer. Parent payload amplitudes MUST NOT select, learn, tune, or alter a kernel coefficient, cutoff or width, normalization, transfer qualification, edge method, support rule, threshold, or other plan fact. No amplitude-thresholded, source-sensitive, spectral-selection, or otherwise data-derived method is admitted. J_full nevertheless MUST use the declared immutable parent facts named in Section 2. Response perturbations, covariance draws, noise realizations, NOI members, and other compatible companions reuse that resolved selector and MUST NOT re-resolve it from their own values.

5. Same-grid finite operator and identity layers

The earlier undifferentiated S_in terminology is replaced by S_parent_fact for typed parent facts and D_m for the numerical function domain. For output row p,

$$y_p = \sum_{q \in D_m} L[p,q] m_q, \quad \text{for } p \in S_{out}, \\ y : S_{out} \rightarrow R.$$

Only entries required by the exact scientific operator may contribute. The operator MUST establish q in D_m before evaluating m_q.

One SCI-FLT-FIXED product applies one exact resolved sampled convolution. A plan may supply a final kernel constructed elsewhere, but the FLT product makes no claim about intermediate transformations. Ordered composition, multiple convolution application, and implicit operator reordering are not admitted in base v0.1. In particular, no equality between a sequence with intermediate selectors and one final selected convolution is presumed.

The FLT-OPERATOR record contains two exact identity layers.

The scientific operator identity binds all of the following:

- exact input D_m and output S_out scientific domains and their relation to S_parent_fact;
- identical parent and output WCS, frame, topology, metric, shape, indexing, and pixel-area convention, apart from restriction to S_out;

- operator family, version, parameter set, canonical coordinate-domain offset-to-coefficient relation, exact coefficient values and units, and scientific-operator digest;
- `K_geom_science`, `K_nonzero`, `K_req`, coefficient coordinate domain, orientation, handedness, center, extent, even or odd tie convention, phase, subpixel convention, coordinate-domain method, and finite scientific support sets;
- normalization and every qualified transfer fact;
- full-footprint row selection and unavailable-row cause rules;
- response, covariance, mode, influence, support, and validity states; and
- requested through realized lifecycle generation, failure, and provenance.

Its digest MUST be independent of dense, sparse, cropped, padded, compressed, container, field-order, byte-order, or other serialization choices.

The separate representation identity binds `K_store`, dense or sparse and cropped or padded encoding, field ordering, byte serialization, compression/container representation, representation digest, and representation generation. A representation-only change MAY create a new representation artifact or representation generation, but it MUST retain the same scientific operator identity, FLT product identity, and scientific transformation and product generation. It MUST NOT alter `S_out`, arithmetic, response, covariance, influence, lifecycle, or any scientific claim. A change to the canonical scientific coefficient map or any other scientific operator fact creates a new scientific transformation and product generation.

Reprojection, resampling, mosaicking, and deconvolution are not same-grid SCI-FLT-FIXED methods. FFT, direct, separable, cached, or threaded evaluation is scientifically immaterial only when it realizes the identical declared finite operator under a preregistered numerical-comparison policy.

6. Fixed convolution and low-pass qualification

For the ordinary exact finite offset representation, fixed convolution is

$$\begin{aligned} (L_{\text{Theta } m})_p &= \text{sum over } r \text{ in } K_{\text{nonzero}} \text{ of } k_{\text{Theta}}(r) m_{(p-r)} \\ &= \text{sum over } r \text{ in } K_{\text{req}} \text{ of } k_{\text{Theta}}(r) m_{(p-r)}. \end{aligned}$$

The sampled kernel, rather than a continuous ideal or family name, constructs the scientific operator. The scientific owner adopts this exact disposition:

- `K_geom_science`: exact scientific geometric footprint, independent of serialization;
- `K_store`: storage or serialization footprint, scientifically nonauthoritative;
- `K_nonzero`: offsets whose canonical coefficient is exactly nonzero, with exact zero decided only from that canonical representation; and
- `K_req` = `K_nonzero` for the ordinary fixed-convolution method.

`K_geom_science` is a representation-invariant geometric description, not the arithmetic dependency set. `K_store` is serialization only. An exact-zero coefficient contributes no arithmetic term, requires no parent payload, creates no influence or covariance contribution, cannot cause row exclusion, and is never classified through a floating threshold. The ordinary method MUST NOT evaluate or dereference a missing, unavailable, or non-finite parent payload at an exact-zero coefficient.

Dense, sparse, cropped, or zero-padded storage of one scientific kernel MUST NOT change `K_geom_science`, `K_nonzero`, `K_req`, `S_out`, response, covariance, scientific operator identity, FLT product identity, or scientific generation. Such encodings have distinct representation identities when their bound serialization facts differ. A floating tolerance MUST NOT classify exact zero. A zero-valued offset may enter `K_req` only under a separately named method with a scientific support reason independent of storage layout. Geometric, storage, nonzero, required-dependency, signed, absolute, and squared support remain distinct objects.

The identity convolution has $K_{\text{req}} = \{\emptyset\}$. Its scientific row domain is the exact admitted finite parent-signal row domain. A nonzero convolution is defined exactly by K_{nonzero} not being empty.

For the exact zero operator,

```

K_nonzero_zero = empty set;
K_req_zero     = empty set;

S_out_zero
  = exact admitted finite parent-signal row domain
    under the zero-operator request and predicates;

y_p = 0, for p in S_out_zero.

```

$S_{\text{out_zero}}$ is constructed independently from the empty arithmetic and required-dependency sets. The zero operator does not describe a parent row domain as a K_{req} domain and does not receive arbitrary storage rows through a vacuous universal predicate. It retains exact parent identity and row support, has local fixed-state derivative zero, and has parent-payload conditional covariance contribution zero. A complete source-domain response or systematic uncertainty may remain typed unavailable.

A convolution declares one exact coordinate-domain method: pixel-index domain, affine tangent-plane angular domain, or another exact identified domain. A fixed shift-invariant pixel kernel is not one spatially uniform angular kernel unless that relation is established over the declared WCS.

An angular-frequency low-pass claim is available only when the resolved plan binds either one exact affine or constant tangent-plane pixel metric, or another exact method proving the stated angular transfer. Every qualified plan content-binds all of the following:

- the exact discrete or Fourier transform sign convention and normalization;
- coordinate and spatial-frequency units;
- zero-frequency origin and ordering;
- positive-frequency, negative-frequency, and Nyquist-frequency treatment;
- the exact frequency sample grid;
- whether the declared response is complex H , amplitude $|H|$, power $|H|^2$, or another exact quantity;
- linear-ratio or decibel attenuation;
- passband, transition, and stopband region geometry;
- complex-phase branch or unwrapping convention;
- anisotropy convention and WCS metric relation; and
- DC gain, finite-grid and edge limitations, sampled kernel, parameter identity, source, and provenance.

For a pixel-index kernel an allowed reference relation is

$$H(\nu) = \sum_{\mathbf{r}} k(\mathbf{r}) \exp[-2\pi i \nu \cdot \mathbf{r}],$$

but the exact adopted convention MUST be content-bound and MUST NOT be inferred from this example. If any required fact is unavailable, fixed-convolution identity may remain available but low-pass qualification is unavailable.

$H(\nu)$ is a possibly complex frequency-domain representation of the real map operator. It does not authorize a complex-valued FLT-SIG. A future complex signal method requires separately named authority for complex signal units, conjugation conventions, Hermitian covariance, and $A \subset A^\dagger$ propagation.

The sampled-kernel or interior translation-invariant transfer is distinct from the complete finite row-restricted operator A_Θ, J . Base v0.1 makes no global Fourier-transfer claim for that complete finite operator unless an exact theorem bound to its finite domain and selector establishes one.

7. Full-footprint scientific output domain

The sole v0.1 scientific output-row method is

```
S_out = {p: for every r in K_req,
            p-r is in S_parent_fact,
            p-r is admitted for this exact FLT use,
            p-r is in D_m,
            and every required predicate passes}.
```

y : S_out -> R.

K_req, not K_store or K_geom_science, governs ordinary scientific admission. Structural finite/non-finite screening uses only authorized required parent locations; an exact-zero geometric offset is not a required location and its payload MUST NOT be evaluated or dereferenced. Rows outside S_out are scientifically unavailable, not zero. A stored array may preserve parent shape and WCS only if every unavailable row carries its typed cause and remains outside the scientific numerical function. FLT MUST establish p-r in D_m before evaluating m_(p-r).

For a requested nonzero convolution that resolves and applies but has S_out = empty set, the exact application state is applied_no_scientific_output_support. The plan, operator, parent, causes, and application evidence are preserved. The ordinary base-signal publication action is not_produced with exact cause no_full_footprint_output_rows. This is not disabled, failed execution, the zero operator, or a successful empty signal product. Identity retains its separately declared support rule; zero uses independently constructed S_out_zero.

Base v0.1 admits no boundary extension, periodic wrapping, truncated convolution, support-conditioned renormalization, inpainting, reflection, clamping, mirroring, edge completion, padding-based admission, or replacement of a missing or non-finite value. Each would require a separately named and versioned successor method.

Numerical computability, complete kernel support, FLT input admission, FLT-local output validity, confidence, and downstream eligibility are distinct states.

8. Quantity, units, beam, response, and modes

FLT-SIG is the transformed parent-map quantity. Its units are derived from the exact parent units and operator coefficient units. For a MAP parent, the originating mJy/beam nominal-beam and calibration lineage remain identified; the output is not relabeled as mJy/filtered-beam.

Kernel normalization is an explicit convention, such as unit signed sum or DC gain, unit peak, unit angular integral, unit L2 norm, or another fully stated convention. No convention is inferred from a name. Numerical unit preservation does not establish calibration, point-source peak preservation, integrated flux, extended-source fidelity, target PSF, or a new beam convention.

Response families are separately typed and none may silently stand for another:

- Fixed-state linear parent response:

$$R_{out}^{fixed} = A_{Theta,J} R_{parent}^{fixed};$$

- An already realized parent-grid response companion receives A_Theta,J exactly once.
- A parent full-procedure finite difference with FLT fixed follows:

$$\Delta y_{parent-FP} = A_{Theta,J} \Delta m_{parent-FP},$$

Baseline and perturbed parent-procedure products MUST provide one exact compatible parent-grid difference on the frozen FLT domain. If row membership, availability, WCS, quantity, or support differs so the difference is not defined on the frozen required footprint, J_full is not re-resolved, the parent

state-change record is retained, and the affected transformed full-procedure response is unavailable. Equal array shape alone does not establish a valid procedure difference.

- An FLT re-resolved procedure response is outside SCI-FLT-FIXED when kernel, cutoff, normalization, support, selector, edge state, or method is re-resolved.

Every admitted response uses the identical frozen $A_{\text{Theta},J}$, centering, orientation, phase, support, edge, missing-data, row-domain, and normalization rules as the signal. If its required source basis, domain, or compatible parent-response identity is unavailable, the requested FLT-RSP family is typed unavailable. The kernel alone is not the complete source response or PSF.

For the exact zero operator, the local fixed-state derivative and the parent-payload conditional covariance contribution are exactly zero on its declared scientific row domain. A complete source-domain response may still be unavailable when its source basis, domain, or parent-response identity is unavailable. Total systematic uncertainty remains separately typed. Local zero response is neither hidden nor promoted into an unsupported complete sky-response claim.

FLT-TRANSFER records the exact local sampled spatial or Fourier transfer on the declared finite-grid domain where scientifically defined, or typed unavailability. FLT-MODE records exact null modes, invariant modes, attenuation statements or bounds, and sampled phase where defined. Local operator modes are not automatically upstream sky-to-output modes.

FLT-INFLUENCE is the exact coefficient and support relation between parent and output rows. It is not physical exposure. Filtering creates no new physical exposure claim. FLT-EXPOSURE-LINEAGE records the exact parent exposure-product identity or typed absence and states that FLT creates no physical exposure. Convolution of an exposure plane is not authorized as the physical exposure of filtered signal.

9. Deterministic covariance and NOI boundary

For an available compatible declared parent covariance C_{parent} ,

$$C_{\text{out}} = A_{\text{Theta},J} C_{\text{parent}} \text{transpose}(A_{\text{Theta},J}).$$

Parent stochastic authority and output representation are separate axes. Parent authority is exactly one of complete covariance, explicit independent-diagonal model, marginal variances only, structured or partial model, or unavailable. Output representation is exactly one of complete matrix, exact linear or operator representation, structured representation, marginal plane, summary only, or unavailable.

Every result states the parent model on which it is conditional, output representation, domain and ordering, rank and null space, omitted terms, supported operations, and selection, kernel, beam, WCS, calibration, and model uncertainty exclusions. A full matrix propagated from an explicitly declared independent-diagonal parent model is complete relative to that conditional model; it does not imply that unknown real parent cross terms are zero.

The exact authority and representation compatibility table is:

Parent authority	Authorized FLT covariance result
complete covariance	exact $A C A^T$ on the complete domain in any mathematically exact declared representation
independent-diagonal model	full covariance relative to that exact model, including induced off-diagonal terms
marginal variances only	exact conditional marginal for a row with exactly one nonzero parent coefficient; otherwise unavailable or explicitly partial
structured or partial model	only operations proved exact for that exact representation and domain
unavailable	unavailable, except separately stated local zero-operator parent-payload facts

Marginal variances alone do not imply independence. For an output row with exactly one nonzero parent coefficient A_{ij} , marginal-only parent authority is sufficient for that row's conditional marginal:

$$\text{Var}(y_i) = A_{ij}^2 \text{Var}(m_j).$$

When a row mixes two or more parent variables, marginal-only authority does not determine its exact output marginal. For a general parent covariance,

$$\begin{aligned} \text{Var}(y_i) = & \sum_j A_{ij}^2 \text{Var}(m_j) \\ & + 2 \sum_{j < k} \text{of } A_{ij} A_{ik} \text{Cov}(m_j, m_k). \end{aligned}$$

A separately named diagonal-contribution diagnostic may report the first sum, but it MUST NOT be called variance, covariance, uncertainty, or precision.

For an explicitly diagonal independent parent covariance with marginal variances V_j ,

$$\begin{aligned} A_{ij} &= (J_{\text{full}} L_{\text{Theta}})_{ij}, \\ \text{Var}(y_i) &= \sum_j A_{ij}^2 V_j, \\ \text{Cov}(y_i, y_l) &= \sum_j A_{ij} A_{lj} V_j. \end{aligned}$$

The output generally has off-diagonal covariance. A marginal variance plane is not full covariance and does not authorize independent-pixel multi-pixel inference. One exact row marginal authorizes neither cross-row covariance nor independence. The zero operator retains its separately typed zero parent-payload covariance contribution.

SCI-NOI, not SCI-FLT-FIXED, owns empirical uncertainty, empirical covariance, conditional inverse scale, standardized signal, and significance inference. For every compatible admitted NOI member M_b , exact fixed-state parity is

$$M_{b,\text{out}} = A_{\text{Theta},J} M_b.$$

The real parent and every member use the identical frozen $A_{\text{Theta},J}$, parameter set, grid, support, edge rule, row domain, and lifecycle generation. J_{full} is never re-resolved per member. Filtering a variance, standard deviation, precision, reciprocal, weight, standardized map, or significance field is not this operation. Per-member selection or re-resolution is a different inference-bearing method and cannot enter the fixed-state ensemble.

Parameter, kernel, cutoff, beam, WCS, selection, model, and calibration uncertainty remain separate from covariance conditional on fixed $A_{\text{Theta},J}$ unless a separately supplied typed model includes them.

10. Atomic product and lifecycle

Every realized bundle contains these role records, including explicit unavailable states where allowed:

- FLT-PARENT;
- FLT-PLAN;
- FLT-OPERATOR;
- FLT-SIG;
- FLT-UNIT-BEAM;
- FLT-TRANSFER;
- FLT-RSP;
- FLT-MODE;
- FLT-INFLUENCE;
- FLT-EXPOSURE-LINEAGE;
- FLT-SUP;
- FLT-VALID;
- FLT-COV-FORMAL;

- FLT-NOI-COMPATIBILITY; and
- FLT-LINEAGE.

A missing required record is not an atomic bundle. An honestly unavailable response or covariance record may satisfy the base role only where the role permits absence; it cannot satisfy a response-qualified or covariance-qualified request.

FLT-NOI-COMPATIBILITY is immutable FLT state, never the NOI product. It binds the exact FLT product, operator, and row-domain identities; exact FLT-to-NOI boundary and profile compatibility; fixed-state transformation semantics; request state known at FLT publication; and typed compatibility or unavailability. It contains no identity of a future NOI product. A later NOI product references the immutable FLT parent. A recorded not_requested_at_FLT_publication state is historical provenance and does not prohibit a later independently requested SCI-NOI child. That child owns its own request, applicability, eligibility, realization, generation, and failure, and still requires exact compatible FLT boundary and profile state. Any optional reverse SCI-FLT-FIXED_TO_SCI-NOI-RELATION is a separately versioned artifact outside FLT atomic completion and MUST NOT mutate the FLT bundle.

Publication policy consumes one exact complete_publication_disposition_candidate with exactly one of two variants:

- product_candidate, containing the complete atomic FLT product bundle; or
- no_output_support_candidate, containing the exact request, parent, plan, scientific operator and representation identities, K_req, attempted output-domain construction, proof that S_out is empty, exact row/cause accounting, application generation, applied_no_scientific_output_support state, and prescribed publication cause.

The no_output_support_candidate contains no realized FLT-SIG and is not an atomic FLT product bundle. The publication lifecycle is

```
requested
-> effective
-> resolved
-> applied
-> complete_publication_disposition_candidate
    (product_candidate | no_output_support_candidate)
-> publication_decision
-> realized | failed | not_produced.
```

For a requested nonzero convolution with empty S_out, the branch is

```
applied
-> applied_no_scientific_output_support
-> complete_publication_disposition_candidate
    (no_output_support_candidate)
-> publication_decision
-> not_produced (cause: no_full_footprint_output_rows).
```

SCI-FLT-FIXED:output_publication@1 evaluates the complete publication disposition candidate, not a pre-existing realized bundle. For the named publication use of a requested nonzero convolution with empty S_out, the exact axes are request = requested, applicability = applicable, eligibility = ineligible, realization = not_produced, and cause = no_full_footprint_output_rows. The earlier input/application use MAY have been eligible and successfully applied. A successful publication action for a product_candidate creates realized_identity, realized_zero, or realized as applicable. Disabled is not_produced. Failure of a required transformation, candidate, or publication step propagates and creates no complete product. unavailable remains a typed route or role state rather than a synonym for disabled, failed, zero, or not produced. superseded records immutable succession after a realized generation exists.

Any change to parent, request or effective purpose, scientific operator, canonical coefficient map, parameter, transfer qualification, normalization, WCS, grid, scientific row domain, support, validity, response or covariance role, lifecycle, or failure policy creates a new immutable scientific transformation and product generation. A representation-only change creates at most a new representation artifact or generation and MUST NOT create a new scientific transformation, FLT product identity, or scientific generation. A later NOI attachment is a separate immutable companion and does not mutate FLT or the parent.

11. Typed FLT policy objects and VAL boundary

The draft defines three distinct policy domains:

- SCI-FLT-FIXED:input_bundle_admission@1 evaluates one request, one complete parent bundle, and one exact resolved FLT plan;
- SCI-FLT-FIXED:input_parent_row_admission@1 evaluates each exact parent row for the named FLT use; and
- SCI-FLT-FIXED:output_publication@1 evaluates one exact complete_publication_disposition_candidate, either product_candidate or no_output_support_candidate, and defines its disposition and prescribed consumer action.

J_full and S_out are deterministic FLT-owned constructions from the parent-row decisions, K_req, S_parent_fact, D_m, and required predicates. Output-row arithmetic support is not a producer fact. Membership in D_m MUST be established before a signal payload is evaluated. SCI-VAL may bind and evaluate an owner-approved profile and produce a decision artifact; it does not convolve data or perform publication. The FLT publisher performs or declines the prescribed action and FLT owns final realization and FLT-local validity.

Every immutable profile binds scientific-policy owner, exact source and boundary versions, request, applicability, eligibility, realization, restrictions, decisive exclusions, exceptions, missing or conflict behavior, lifecycle, and exact consumer action. The exact draft records are part of the bound Stage B source set.

Structural identity or source conflict yields applicability_unknown and decision_unavailable. A known applicable failed restriction yields ineligible. A not-requested or effectively disabled route makes no eligibility proposition and yields not_produced. An eligible transformation failure yields eligible and realization_failed. An eligible product_candidate becomes realized only after successful publication action. An applied nonzero transformation with empty S_out yields applied_no_scientific_output_support and a complete no_output_support_candidate; for that publication use policy records request = requested, applicability = applicable, eligibility = ineligible, realization = not_produced, and cause no_full_footprint_output_rows, and publishes no empty base-signal product.

Companion qualification is request-specific:

- a base signal request requires exact response and covariance state records but permits an honest unavailable state where their role allows it;
- a response-qualified request requires an available exact compatible response;
- a covariance-qualified request requires the exact declared compatible covariance representation; and
- a response-and-covariance-qualified request requires both.

An optional unavailable companion never blocks an otherwise valid base-signal product. It does block the corresponding qualified request.

SCI-VAL may bind and evaluate an immutable owner-approved successor of these profiles. VAL does not author producer facts, FLT policy, arithmetic, or scientific claims. The profile records first drafted at r0.3 and

their current exact source bindings remain drafts, are not owner-approved Registry entries, and create no claim that Registry evaluation occurred.

12. Consumer and ownership boundaries

MAP and JINC own the parent estimand and parent claims. CAL owns absolute calibration, passband and color correction, and calibration covariance. SCI-FLT-FIXED owns the exact local transformation, transformed signal, output unit derivation, composed-response state, local transfer and modes, influence, support and validity, deterministic covariance state, lifecycle, failure, and provenance. It also owns its plan, selector, application, and publication policy even when a final kernel is constructed elsewhere. SCI-NOI owns empirical uncertainty inference and applies but does not choose the exact FLT transformation. SCI-BEAM and future source or mode contracts own physical source, beam, Pointing, and OOF interpretations. SCI-FRUIT owns iterative feedback science.

confidence is not a generic FLT scalar. Where the applicable upstream boundary supplies a named confidence or quality state, FLT carries its exact typed identity and meaning as parent state and may reference it in a named row predicate. Otherwise FLT confidence is `not_defined`; it must not be inferred from finiteness, support, covariance, a weight, or downstream eligibility.

The availability of an SCI-FLT-FIXED product authorizes no generic Beammap, Pointing, OOF, source-fit, catalog, NOI, or FRUIT use. Each consumer owns an exact use policy.

13. Numerical-conformance policy boundary

Future finite-precision implementation evidence must be evaluated under one preregistered immutable numerical-comparison policy. Before observing a candidate result, that policy binds an independent exact or high-precision oracle; absolute, relative, and near-zero behavior; signed-cancellation and zero-sum cases; conditioning and operation-count dependence; covariance comparison; sequential and parallel agreement; overflow, underflow, and non-finite handling; simultaneous row-level comparison; lifecycle; and provenance. Bounds must not be changed after observing a failure.

The policy preserves exact operator coefficients and scientific identity. It authors no filter science and supplies no implementation-conformity, validation, numerical-adequacy, or performance claim. The current draft policy is bound as a separate Stage B source artifact and is not a preregistered future evidence decision.

14. Exclusions and nonclaims

This contract excludes affine offsets, background or template subtraction, additive correction, reprojection, resampling, mosaicking, deconvolution, boundary extension, truncated or renormalized convolution, missing-value replacement, data-derived kernel or cutoff selection, automatic method selection, Wiener transformation, matched or generalized least-squares template-amplitude estimation, source-learned operation, data-derived mode selection or map-domain destriping, per-member relearning, FLT coaddition, source or mode interpretation, FRUIT recurrence, and RTC timestream filtering.

This Stage B draft makes no implementation-conformity, algorithm-change, validation, calibration, achieved-response, achieved-covariance, numerical adequacy, performance, readiness, scientific-freeze, production, or Unity claim.

15. Normative requirements

SCI-FLT-FIXED-REQ-001 - Package identity

The package SHALL be identified as SCI-FLT-FIXED v0.1 within the SCI-FLT tranche. FLT-FIXED-CONV SHALL be the only numerically admitted base family, with FLT-FIXED-CONV-LOWPASS only a qualified subtype. It SHALL NOT be presented as SCI-FLT-INF, an arbitrary dense linear-operator family, or a generic filter contract.

SCI-FLT-FIXED-REQ-002 - Strict linearity

The transformation MUST be exactly $y = A_{\text{Theta},J} m$, where $A_{\text{Theta},J} = J_{\text{full}} L_{\text{Theta}}$, with no additive term. Strict linearity MUST be conditioned on one exact frozen parent membership and selector, with $m : D_m \rightarrow R$ and $y : S_{\text{out}} \rightarrow R$. $S_{\text{parent_fact}}$, D_m , and S_{out} MUST remain separately typed. Any offset, background, template subtraction, or additive correction MUST be rejected as outside v0.1.

SCI-FLT-FIXED-REQ-003 - Exact parent role

Each transformation MUST bind exactly one complete immutable FLT-PARENT-MAP-OBS, FLT-PARENT-MAP-COADD, or FLT-PARENT-JINC-OBS parent. No role substitution by shape, WCS, name, or payload similarity is permitted.

SCI-FLT-FIXED-REQ-004 - Complete parent identity

The parent binding MUST include all applicable package, revision, product, application, observation or membership, quantity, unit, nominal-beam, WCS, grid, row-domain, support, validity, response, covariance, null, exposure, lifecycle, failure, and provenance facts stated in Section 3. A partial JINC five-role bundle MUST be rejected.

SCI-FLT-FIXED-REQ-005 - Honest upstream availability

An unavailable MAP or JINC numerical parent MUST remain unavailable. Algebra, defaults, approximate identity, or a finite payload MUST NOT manufacture a parent or successor.

SCI-FLT-FIXED-REQ-006 - Parent ordering

Filtering an observation and filtering a coadd MUST create distinct successor identities. SCI-FLT-FIXED MUST NOT coadd or claim filter/coadd commutation without a separately approved exact bounded relation.

SCI-FLT-FIXED-REQ-007 - External fixed resolution

Every operator coefficient, parameter, grid fact, normalization, support rule, and transfer qualification MUST be resolved under one exact FLT-owned plan and frozen before application. Parent payload amplitudes MUST NOT select, learn, tune, or alter any plan fact. The selector MUST be resolved once from only the declared immutable $S_{\text{parent_fact}}$, row membership and admission, typed facts defining D_m , support, and predicates in Section 2 before convolution arithmetic. FLT MUST establish membership in D_m before evaluating a signal payload and MUST NOT re-resolve the selector for response, covariance, noise, or NOI members.

SCI-FLT-FIXED-REQ-008 - Same-grid boundary

The output MUST preserve the exact parent WCS, frame, topology, metric, shape, pixel indexing, and pixel-area convention, apart from scientific-row restriction. Reprojection, resampling, and approximate-WCS joins MUST be rejected.

SCI-FLT-FIXED-REQ-009 - Complete operator identity

The applied operator MUST bind the separate scientific-operator and representation identities in Sections 5 and 6. Scientific identity MUST bind the canonical exact offset-to-coefficient map and units, $K_{\text{geom_science}}$, K_{nonzero} , K_{req} , scientific domains, coordinate and WCS facts, normalization, transfer qualification, edge rule, and all downstream science. Representation identity MUST separately bind K_{store} , encoding, field and byte ordering, container/compression, representation digest, and representation generation. K_{store} MUST be scientifically nonauthoritative; the scientific digest and every scientific support set MUST be representation-invariant. Each coefficient MUST satisfy the real, finite, unit-typed, canonical, content-bound conditions in Section 2.

SCI-FLT-FIXED-REQ-010 - Fixed convolution construction

A FLT-FIXED-CONV method MUST construct the complete finite operator from one exact finite sampled kernel and its declared offset sets, orientation, center, phase, normalization, coordinate-domain method, and coefficient representation. Its ordinary sum MUST range exactly over $K_{\text{nonzero}} = K_{\text{req}}$, never $K_{\text{geom_science}}$ or K_{store} . One product MUST apply this convolution exactly once and MUST make no intermediate transformation or reordered-composition claim.

SCI-FLT-FIXED-REQ-011 - Low-pass qualification

A FLT-FIXED-CONV-LOWPASS claim MUST bind every coordinate, metric, frequency, and transfer fact listed in Section 6 and distinguish sampled-kernel transfer from the complete finite row-restricted operator. If any fact is missing, the low-pass claim MUST be unavailable even when fixed-convolution identity remains available.

SCI-FLT-FIXED-REQ-012 - Full-footprint admission

S_{out} MUST use ordinary-method $K_{\text{req}} = K_{\text{nonzero}}$ and contain exactly those rows for which every scientifically required parent location is in $S_{\text{parent_fact}}$, admitted for the exact FLT use, in D_m , and passing every required predicate. Membership in D_m MUST be established before m is evaluated. Exact-zero coefficient status MUST be read from the canonical representation, never a floating threshold. Storage layout MUST NOT change row admission. An exact-zero offset MUST require no parent payload, MUST create no influence or covariance contribution, MUST NOT remove a row, and MUST NOT cause its payload to be evaluated or dereferenced.

SCI-FLT-FIXED-REQ-013 - Unavailable row semantics

A row outside S_{out} MUST be unavailable rather than zero. Parent-shaped storage MAY be retained only with explicit unavailable-row state and causes.

SCI-FLT-FIXED-REQ-014 - Sole edge method

Boundary extension, wrapping, truncation, support renormalization, inpainting, reflection, clamping, mirroring, padding-based admission, edge completion, and missing or non-finite replacement MUST NOT be used by v0.1.

SCI-FLT-FIXED-REQ-015 - Output quantity and units

FLT-SIG MUST be identified as transformed parent-map amplitude on S_{out} , with output units derived from exact parent and coefficient units. It MUST NOT be relabeled as flux, fitted amplitude, or a response-corrected quantity.

SCI-FLT-FIXED-REQ-016 - Beam and calibration boundary

The originating nominal-beam and calibration lineage MUST be retained. No new filtered-beam, target-PSF, absolute-calibration, peak-preservation, integrated-flux, or extended-source claim may be inferred from normalization or units.

SCI-FLT-FIXED-REQ-017 - Response composition

Each FLT-RSP MUST identify exactly one response family in Section 8. An available compatible fixed-state response, realized parent-grid response, or parent full-procedure finite difference MUST receive the identical frozen $A_{\text{Theta}, J}$ exactly once as specified. A full-procedure difference MUST be defined on one exact compatible frozen parent-grid domain; a membership, availability, WCS, quantity, or support mismatch MUST yield affected-response unavailability without re-resolving J_{full} . FLT re-resolution is outside v0.1. A missing required basis, domain, or parent-response identity MUST yield typed unavailability rather than substitution.

SCI-FLT-FIXED-REQ-018 - Transfer and mode state

FLT-TRANSFER and FLT-MODE MUST publish the exact sampled-kernel or local finite-grid transfer, null, invariant, attenuation, and phase facts where defined, or honest typed unavailability. They MUST NOT be promoted to an unproved global finite-operator, angular-WCS, or whole-chain claim.

SCI-FLT-FIXED-REQ-019 - Influence is not exposure

FLT-INFLUENCE MUST describe the exact parent-to-output coefficient relation and MUST NOT be labeled or interpreted as physical exposure. FLT-EXPOSURE-LINEAGE MUST identify the exact parent exposure product or typed absence and MUST state that FLT creates no physical exposure.

SCI-FLT-FIXED-REQ-020 - Distinct support and validity states

Numerical computability, K_geom_science, K_store, K_nonzero, K_req, complete footprint, FLT bundle and row admission, FLT-local validity, parent validity, named upstream confidence or not_defined, and downstream eligibility MUST remain distinct with exact causes.

SCI-FLT-FIXED-REQ-021 - Deterministic covariance propagation

An available compatible parent covariance MUST be propagated by $C_{out} = A_{\Theta, J} C_{parent} \text{transpose}(A_{\Theta, J})$ on the exact scientific row domain with the identical frozen selector and only under the exact authority and representation compatibility rules in Section 9.

SCI-FLT-FIXED-REQ-022 - Covariance state honesty

FLT-COV-FORMAL MUST separately state parent stochastic authority and output representation using Section 9's two axes, plus conditional model, domain, ordering, rank, null space, omissions, supported operations, and excluded uncertainties. Unknown cross terms MUST NOT be set to zero or interpreted as independence.

SCI-FLT-FIXED-REQ-023 - Induced covariance

For an explicitly diagonal independent parent model, the exact output off-diagonal terms using A_{ij} MUST be retained in any complete-relative-to- that-model covariance claim. Marginal parent variances MUST NOT infer independence or determine an exact output marginal for a row mixing multiple parent variables. A marginal plane MUST NOT be called full covariance, and a diagonal-contribution diagnostic MUST NOT be called variance, covariance, uncertainty, or precision.

SCI-FLT-FIXED-REQ-024 - Empirical uncertainty ownership

SCI-FLT-FIXED MUST NOT infer empirical uncertainty, empirical covariance, conditional inverse scale, standardized signal, or significance. Such an attachment remains SCI-NOI-owned and separate from FLT-COV-FORMAL.

SCI-FLT-FIXED-REQ-025 - Fixed-state NOI parity

Every admitted NOI member for the exact transformed product MUST receive the identical frozen $A_{\Theta, J}$, parameters, grid, support, edge rule, row domain, and lifecycle generation used for the real parent. A member failing a frozen required footprint MUST be unavailable there rather than receiving a new selector.

SCI-FLT-FIXED-REQ-026 - Relearning rejection

Any per-member parameter selection, operator resolution, or relearning MUST be rejected from the fixed-state ensemble and routed to a separately named inference-bearing method.

SCI-FLT-FIXED-REQ-027 - Atomic product roles

A realized product MUST contain every required role record in Section 10, including explicit unavailable companion records where the base role permits absence. Immutable FLT-NOI-COMPATIBILITY, not an SCI-NOI product, future NOI identity, or later-changing attachment state, MUST occupy the FLT relation role. A partial bundle MUST NOT be published.

SCI-FLT-FIXED-REQ-028 - Lifecycle

The lifecycle MUST evaluate publication policy against one exact `complete_publication_disposition_candidate`, either `product_candidate` or `no_output_support_candidate`, before realization and MUST distinguish all states and transitions in Section 10 while preserving exact causes, failures, and immutable generation bindings.

SCI-FLT-FIXED-REQ-029 - Disabled, identity, and zero states

Disabled MUST be not `_produced`. Requested and applied identity and zero operators MUST use their explicit support rules, `product_candidate`, and publication decision to produce distinct realized products and MUST NOT be represented as disabled or unavailable. Zero MUST use `K_nonzero_zero = empty set`, `K_req_zero = empty set`, and independently constructed `S_out_zero`; it MUST NOT acquire rows through a vacuous empty-footprint test.

SCI-FLT-FIXED-REQ-030 - Generation identity

Any change to a scientific operator fact listed in Section 10 MUST create a new immutable scientific transformation and product generation. A representation-only change MAY create a representation artifact or generation but MUST retain the same scientific operator identity, FLT product identity, and scientific generation. A separately realized NOI product or optional reverse relation MUST reference the immutable FLT product, MUST NOT determine FLT completeness, and MUST NOT mutate the FLT product.

SCI-FLT-FIXED-REQ-031 - Failure and fallback

Missing, conflicting, or unavailable state required by the requested identity MUST fail the affected route closed with exact causes. No silent fallback, default, or same-name substitution may retain that requested identity.

SCI-FLT-FIXED-REQ-032 - Input admission policy

The SCI-FLT-FIXED:input_bundle_admission@1 and SCI-FLT-FIXED:input_parent_row_admission@1 draft semantics in Section 11 MUST govern their distinct bundle and exact parent-row domains, typed applicability and eligibility, unavailable decision, exclusions, and fail-closed action until superseded by owner-approved immutable profiles. `J_full` and `S_out` MUST remain deterministic FLT constructions, not profile-authored output-row arithmetic.

SCI-FLT-FIXED-REQ-033 - Output publication policy

The SCI-FLT-FIXED:output_publication@1 draft semantics in Section 11 MUST govern the exact `complete_publication_disposition_candidate`, its `product_candidate` and `no_output_support_candidate` variants, disposition, and prescribed consumer action. A policy or VAL evaluation MUST NOT itself perform publication; the FLT publisher performs or declines that action. Honest absence is permitted only for a role and request qualification that explicitly allows it.

SCI-FLT-FIXED-REQ-034 - VAL boundary

SCI-VAL MAY bind and evaluate an owner-approved immutable policy and produce a decision artifact but MUST NOT author FLT facts or policy, perform the transformation or publication, or convert these unapproved draft profile bytes into a Registry-evaluated or numerical route.

SCI-FLT-FIXED-REQ-035 - Consumer boundary

Product availability MUST NOT authorize generic downstream use. Beammap, Pointing, OOF, source-fit, catalog, NOI, and FRUIT consumers MUST supply their own exact owner-approved use policies.

SCI-FLT-FIXED-REQ-036 - Excluded methods and nonclaims

Every method in Section 14 MUST remain outside v0.1. The package MUST preserve all Stage B nonclaims in Section 14.

SCI-FLT-FIXED-REQ-037 - Frozen-selector conditioning

J_{full} MUST be resolved exactly once before convolution arithmetic from only declared immutable S_{parent_fact} , row membership, exact parent-row admission, typed facts defining D_m , support, and exact required predicates. Membership in D_m MUST be established before m_q is evaluated. Structural screening of declared typed row-state facts MUST NOT evaluate m_q or tune the plan. Response perturbations, covariance draws, noise realizations, and NOI members MUST reuse the resulting $A_{Theta,J}$; selection and support uncertainty MUST remain excluded unless separately supplied as typed uncertainty.

SCI-FLT-FIXED-REQ-038 - Required-dependency support

Every kernel MUST distinguish representation-invariant $K_{geom_science}$, nonauthoritative K_{store} , $K_{nonzero}$, and K_{req} and bind their exact relations. The ordinary method MUST use $K_{req} = K_{nonzero}$; identity uses $K_{req} = \{0\}$; and the zero operator uses $K_{nonzero_zero} = \text{empty_set}$, $K_{req_zero} = \text{empty_set}$, and independently constructed S_{out_zero} equal to the exact admitted finite parent-signal row domain under its request and predicates. A parent row domain MUST NOT be described as a K_{req} domain. An exact-zero ordinary coefficient contributes no arithmetic dependency, influence, covariance, or row exclusion and its parent payload MUST NOT be dereferenced. A zero-valued required offset requires a separately named scientific method independent of storage.

SCI-FLT-FIXED-REQ-039 - Typed policy and companion qualification

All three policy objects in Section 11 MUST bind their exact domains and VAL- congruent state semantics. Base, response-qualified, covariance-qualified, and jointly qualified requests MUST apply their distinct companion requirements; an optional unavailable companion MUST NOT block a valid base signal.

SCI-FLT-FIXED-REQ-040 - Response-family separation

Fixed-state linear response, already realized parent-grid response, parent full-procedure finite difference with FLT fixed, and FLT re-resolved procedure response MUST remain separately identified. No response family or local zero derivative may be promoted into another or into an unsupported complete source-domain claim.

SCI-FLT-FIXED-REQ-041 - Single-convolution composition

One SCI-FLT-FIXED v0.1 product MUST apply one exact resolved sampled convolution exactly once. Intermediate transformations, multiple application, selector collapse, or operator reordering MUST NOT be claimed by the product.

SCI-FLT-FIXED-REQ-042 - Numerical-comparison policy

Any future finite-precision conformance evidence MUST preregister every comparison-policy field in Section 13 before candidate results are observed. Comparison bounds MUST NOT change after a failure, and the policy MUST NOT authorize filter science or claim validation, adequacy, or performance.

SCI-FLT-FIXED-REQ-043 - Exposure lineage

Every complete candidate MUST carry FLT-EXPOSURE-LINEAGE with the exact parent exposure identity or typed absence. FLT MUST create no physical exposure, MUST keep influence distinct, and MUST NOT authorize a convolved exposure plane as filtered-signal physical exposure.

SCI-FLT-FIXED-REQ-044 - External-resolution ownership

"Externally resolved" MUST mean complete before application under one exact FLT-owned plan. It MUST NOT transfer FLT policy, transformation, selector, application, or publication ownership to an unnamed producer.

SCI-FLT-FIXED-REQ-045 - Exact parent signal role

Each parent role MUST bind the exact signal vector in Section 3. Only that signal role may receive L_Theta as FLT-SIG. JINC numerator, signed normalization, quadratic accumulator, and coefficient-squared temporal accounting; MAP exposure; support; validity; covariance; and response MUST NOT be transformed as FLT-SIG by the ordinary method.

SCI-FLT-FIXED-REQ-046 - Covariance authority compatibility

Every covariance result MUST follow the Section 9 authority/representation table. Complete, independent-diagonal, marginal-only, structured or partial, and unavailable parent authorities MUST remain distinct. Unknown cross terms MUST keep any marginal for a row mixing multiple parent variables, and any unsupported cross-row covariance, unavailable or explicitly partial. A row with exactly one nonzero parent coefficient MAY publish its exact conditional marginal without implying independence.

SCI-FLT-FIXED-REQ-047 - Immutable NOI compatibility

FLT-NOI-COMPATIBILITY MUST contain only FLT identity, operator and row-domain identity, boundary/profile compatibility, fixed-state semantics, publication- time request state, and typed compatibility or unavailability. It MUST NOT contain a future NOI identity. A later NOI child or reverse relation MUST NOT mutate FLT. `not_requested_at_FLT_publication` MUST be historical provenance only and MUST NOT prohibit a later independently requested compatible SCI-NOI child with its own lifecycle.

SCI-FLT-FIXED-REQ-048 - Policy actor separation

Input profiles MUST evaluate complete bundles and exact parent rows. FLT MUST construct `J_full` and `S_out`. Publication policy MUST evaluate either exact candidate variant and define a disposition and prescribed action; VAL MAY produce a decision artifact; only the FLT publisher may perform or decline publication and establish FLT realization and local validity.

SCI-FLT-FIXED-REQ-049 - Low-pass transform convention

Every low-pass-qualified plan MUST content-bind every transform, frequency, quantity, attenuation, region, phase, anisotropy, and WCS-metric convention in Section 6. A different convention, even with complete metadata, MUST fail the qualification.

SCI-FLT-FIXED-REQ-050 - Full-procedure response domain

A transformed parent full-procedure response MUST use one exact compatible baseline/perturbed difference on the frozen FLT domain. Membership, availability, WCS, quantity, or support incompatibility MUST retain the parent state-change record, MUST NOT re-resolve `J_full`, and MUST make affected response rows unavailable.

SCI-FLT-FIXED-REQ-051 - Consolidated authority preflight

`AUTHORITY_MANIFEST.json`, the proposed-freeze authority manifest, MUST bind the complete Stage A packet, all owner directives, every Stage B scientific and policy source, exact build and verification records, reports, and rendered PDFs by path, bytes, and SHA-256, plus scientific/process/representation role, authority state, compatibility or supersession state, and generated-view relation. Any unreproducible dependency MUST route its dependent claim to typed unavailability rather than freeze disposition.

SCI-FLT-FIXED-REQ-052 - Real scalar and coefficient admissibility

The base functions MUST be $m : D_m \rightarrow R$ and $y : S_{out} \rightarrow R$, with S_{parent_fact} , D_m , and S_{out} separately typed; $k_{Theta}(r)$, L_{Theta} , and $A_{Theta,J}$ MUST be real-valued. Every sampled coefficient MUST be finite, real, unit-typed, canonically exactly represented, and content-bound before application. A missing, non-finite, complex, unrepresentable, or conflicting coefficient MUST make plan resolution unavailable and MUST NOT be repaired by numerical-comparison policy. Complex $H(\nu)$ MUST remain only a representation of the real map operator.

SCI-FLT-FIXED-REQ-053 - Empty scientific output support

For a requested nonzero convolution that resolves and applies with $S_{out} = \text{empty_set}$, FLT MUST record `applied_no_scientific_output_support`, preserve its bound evidence in a complete `no_output_support_candidate`, and prescribe the publication-use axes request = requested, applicability = applicable, eligibility = ineligible, realization = not_produced, and cause `no_full_footprint_output_rows`. The candidate MUST contain no realized FLT-SIG and MUST NOT be an atomic bundle. FLT MUST NOT publish a successful empty signal product or relabel the state as not requested, disabled, failed execution, decision unavailable, identity, or zero.

16. Falsifiable predictions

Each prediction is conditional on an exact admitted parent and fully resolved operator unless the prediction explicitly tests unavailability.

SCI-FLT-FIXED-PRED-001 - Identity operator

For the exact identity operator, FLT-SIG equals the parent signal on S_{out} . An available compatible response and covariance are unchanged on that domain. $K_{req} = \{0\}$ and S_{out} equals the exact admitted finite parent-row domain. The result is `realized_identity` only after successful publication, not disabled and not an unparented copy.

SCI-FLT-FIXED-PRED-002 - Zero operator

For the exact zero operator, $K_{nonzero_zero}$ and K_{req_zero} are empty, while S_{out_zero} is independently the exact admitted finite parent-signal row domain under the request and predicates. Every FLT-SIG value on S_{out_zero} is zero; the local fixed-state derivative and parent-payload conditional covariance contribution are zero there. No arbitrary storage row is admitted by vacuity. A missing complete source basis or parent response remains typed unavailable, and total systematic uncertainty remains separate. The product is `realized_zero` only after publication, not disabled, invalid, or evidence of precision.

SCI-FLT-FIXED-PRED-003 - Input scaling

For any finite scalar a , applying the same frozen $A_{Theta,J}$ to $a \cdot m$ produces $a \cdot y$. Changing parent membership or re-resolving J_{full} is outside this conditional-linearity comparison. Failure on the fixed domain falsifies strict linearity.

SCI-FLT-FIXED-PRED-004 - Constant input and DC gain

For constant admitted input $m_q = c$, every full-footprint convolution row is c times the exact signed kernel sum, equivalently c times the declared DC gain. Constant preservation occurs exactly when that gain is one; no other normalization implies it.

SCI-FLT-FIXED-PRED-005 - Impulse response

An admitted single-pixel unit impulse produces the exact sampled kernel shifted according to the declared center, orientation, handedness, phase, and indexing, with arithmetic terms only at exact $K_{nonzero} = K_{req}$ offsets, then restricted by S_{out} . Any implicit recentering, interpolation, reversal, periodic copy, or geometric exact-zero term falsifies the operator identity.

SCI-FLT-FIXED-PRED-006 - Parent-response composition

Applying the transformation to an exact compatible fixed-state response, already realized response companion, or parent full-procedure difference produces exactly its separately identified $A_{\text{Theta}, J}$ composition. A kernel-only surrogate, a second application, a re-resolved selector, or a response using different edge, phase, or normalization rules fails.

SCI-FLT-FIXED-PRED-007 - Signed kernel

A signed kernel follows its exact signed coefficients, including cancellation. Its geometric, signed, absolute, and squared support summaries remain unequal when the coefficients make them unequal; none may substitute for another.

SCI-FLT-FIXED-PRED-008 - Zero-sum kernel

For a complete constant input and exact zero signed-sum kernel, every admitted output row is zero. This nulling does not by itself make the row unavailable, uncertain, source-free, or statistically significant.

SCI-FLT-FIXED-PRED-009 - Full-footprint boundary

For every output row whose exact K_{req} footprint lies in $S_{\text{parent_fact}}$, is admitted, lies in D_m , and passes every predicate, the row is in S_{out} . Removing a required location from $S_{\text{parent_fact}}$ or D_m , or failing its admission or predicate, removes every dependent output row with an exact cause; m_q is never evaluated before q in D_m is established. Dense and sparse representations with distinct representation identities but the identical canonical coefficient map produce the identical scientific operator identity, FLT product identity, K_{req} , and S_{out} . An unavailable or non-finite payload at an exact-zero geometric offset leaves the row and value unchanged; making that coefficient exactly nonzero activates the dependency and removes the row.

SCI-FLT-FIXED-PRED-010 - Deferred edge methods

Any result that uses extension, wrapping, truncation, local support renormalization, inpainting, reflection, clamp, mirror, padding-based admission, edge completion, or value replacement is rejected as a v0.1 product even when it is finite or preserves a constant.

SCI-FLT-FIXED-PRED-011 - Missing and non-finite input

No output row depending on a required location outside $S_{\text{parent_fact}}$, outside D_m , not admitted, or failing a required predicate belongs to S_{out} . Such a row is unavailable rather than zero and records the applicable cause. A stored missing, unavailable, or non-finite payload remains a typed fact and is never evaluated as an element of m .

SCI-FLT-FIXED-PRED-012 - Complete covariance transform

For an exact available compatible C_{parent} , the published complete covariance equals $A_{\text{Theta}, J} C_{\text{parent}}^{\text{transpose}}(A_{\text{Theta}, J})$ on the declared row ordering. A different domain, ordering, or omitted required term fails the complete- covariance claim.

SCI-FLT-FIXED-PRED-013 - Off-diagonal covariance from diagonal input

For an explicitly independent-diagonal parent model, overlapping nonzero output rows produce the exact cross term $\sum_j A_{ij} A_{lj} V_j$. Publishing only marginals cannot pass a complete-relative-to-that-model covariance check when this term is nonzero, and that conditional completeness says nothing about unknown real parent cross terms.

SCI-FLT-FIXED-PRED-014 - Unavailable parent companion

For a base signal request, unavailable parent response or covariance yields an honest unavailable state record without blocking an otherwise complete signal. The same state fails a corresponding response-qualified or covariance-qualified request. A zero array, diagonal guess, weight, denominator, or kernel-only surrogate fails.

SCI-FLT-FIXED-PRED-015 - WCS or grid mismatch

Any mismatch in parent and output WCS, frame, topology, metric, shape, pixel indexing, or pixel-area convention makes the same-grid route unavailable. Approximate equality or successful numerical resampling does not pass.

SCI-FLT-FIXED-PRED-016 - Observation and coadd identity

A filtered observation, filtered MAP coadd, and filtered JINC observation have distinct parent and successor identities even if their arrays are numerically equal. No coadd of filtered observations is an SCI-FLT-FIXED v0.1 product.

SCI-FLT-FIXED-PRED-017 - Exact NOI parity

For each compatible admitted NOI member, applying the exact frozen signal operator produces A_{Theta} , J_{M_b} on the identical row domain. A member missing one frozen required footprint is unavailable there; a different coefficient, parameter, grid, support, edge rule, generation, or selector makes the route unavailable.

SCI-FLT-FIXED-PRED-018 - Per-member re-resolution

If any NOI member selects or re-resolves a kernel, cutoff, support, threshold, edge state, or other parameter, that member is rejected from the fixed-state ensemble rather than mixed with it.

SCI-FLT-FIXED-PRED-019 - Disabled, identity, zero, and failure

A disabled route is not produced; identity and zero each pass through a complete product_candidate and publication decision before emitting their separately parented realized products; and a required failure emits no complete product and propagates its cause. Collapsing any pair fails the lifecycle contract.

SCI-FLT-FIXED-PRED-020 - Upstream unavailable parent

At the launch state recorded by the admitted boundaries, an attempted ordinary numerical MAP or TolTEC JINC transformation remains unavailable unless a separately authorized upstream successor supplies every named gate. The FLT contract alone cannot make that route numerical.

SCI-FLT-FIXED-PRED-021 - Low-pass claim completeness

Removing any required low-pass transfer fact from an otherwise complete fixed convolution makes only the low-pass qualification unavailable. Retaining a low-pass label with an incomplete transfer specification fails the contract.

SCI-FLT-FIXED-PRED-022 - Exact-zero and storage invariance

For the ordinary method, an exact-zero canonical coefficient is absent from K_{nonzero} and K_{req} . Dense and sparse encodings bind different representation identities and generations when their representation facts differ, while preserving the identical scientific operator identity, FLT product identity, scientific generation, $K_{\text{geom_science}}$, K_{nonzero} , K_{req} , S_{out} , arithmetic, influence, covariance, response, and claim. Its parent payload is not evaluated or dereferenced. Making that coefficient any exact nonzero value changes the canonical scientific map and activates a new scientific operator identity, arithmetic, row dependency, and scientific generation. A floating threshold or storage layout that changes this result falsifies the identity separation.

SCI-FLT-FIXED-PRED-023 - Zero-operator row support

The exact zero operator has $K_{\text{nonzero_zero}} = \text{empty set}$ and $K_{\text{req_zero}} = \text{empty set}$ and produces rows exactly on independently constructed $S_{\text{out_zero}}$, the admitted finite parent-signal row domain under its request and predicates. An empty required footprint that admits storage rows outside $S_{\text{out_zero}}$, or a zero array with no parent identity and row support, fails the contract. Its empty arithmetic sum creates no parent-payload covariance contribution and is not the empty S_{out} state of a nonzero convolution, which by definition has nonempty K_{nonzero} .

SCI-FLT-FIXED-PRED-024 - Independent low-pass transfer check

For a qualified low-pass subtype, an independent exact sampled-transfer evaluation using the identical content-bound transform sign, normalization, units, origin, ordering, signed/Nyquist treatment, frequency grid, response quantity, attenuation units, region geometry, phase, anisotropy, and WCS metric reproduces the declared DC gain and transfer. Complete metadata under a different convention fails. PRED-021 metadata completeness without this numerical agreement is insufficient.

SCI-FLT-FIXED-PRED-025 - Non-signal role rejection

An ordinary-method attempt to transform JINC numerator, signed normalization, quadratic accumulator, coefficient-squared temporal accounting, MAP exposure, support, validity, covariance, or response as FLT-SIG is rejected. The same objects may enter only their separately typed facts or compositions.

SCI-FLT-FIXED-PRED-026 - Covariance cross-term sensitivity

For two compatible parent covariance matrices with identical diagonals and different cross terms, any filtered output row that mixes the affected parent variables has different variance when the corresponding weighted cross term is nonzero. A marginal-only parent selects neither result and leaves the exact output marginal unavailable or explicitly partial. In contrast, a row with exactly one nonzero coefficient has conditional marginal $A_{ij}^2 \text{Var}(m_j)$ from marginal-only authority; this authorizes no cross-row covariance or independence. The exact-zero row keeps its separately typed zero parent-payload contribution.

SCI-FLT-FIXED-PRED-027 - NOI child nonmutation

Realizing, replacing, superseding, or removing a later SCI-NOI child or an optional reverse relation leaves the immutable FLT bundle bytes, completeness, realization, and FLT-NOI-COMPATIBILITY unchanged. Any mutation fails the atomic product contract. A `not_requested_at_FLT_publication` record does not block a later independently requested compatible child; that child owns its own request, applicability, eligibility, realization, generation, and failure.

SCI-FLT-FIXED-PRED-028 - Full-procedure domain incompatibility

If baseline and perturbed parent-procedure products share array shape but differ in row membership, availability, WCS, quantity, or required support, the affected transformed full-procedure response is unavailable, `J_full` remains frozen, and the parent state-change record remains visible. Producing a numerical difference by shape alone fails.

SCI-FLT-FIXED-PRED-029 - Empty scientific output support

For a requested nonzero convolution wider than its parent, or otherwise having no complete admitted footprint, application yields `S_out = empty set` and `applied_no_scientific_output_support`. It creates a complete `no_output_support_candidate` containing the exact attempted-domain proof and evidence but no realized FLT-SIG and no atomic bundle. For the publication use, the exact axes are requested, applicable, ineligible, and not produced, with cause `no_full_footprint_output_rows`. A not-requested, disabled, execution-failed, decision-unavailable, zero-operator, or realized empty-signal result fails.

SCI-FLT-FIXED-PRED-030 - Invalid coefficient rejection

A kernel containing any missing, non-finite, complex, unrepresentable, or conflicting coefficient makes plan resolution unavailable before application. No numerical-comparison tolerance repairs it. A kernel with all coefficients finite, real, unit-typed, canonically represented, and content-bound passes this coefficient-admissibility fixture without thereby making any low-pass, response, covariance, or implementation claim.